

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

MEDICAL : NURTURE PHASE-I & II

TEST # 02

DATE : 27-05-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry and Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

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PART-I [PHYSICS]

- | | |
|---|---|
| <p>1. Which of the following statements are correct?
 (1) $\sin^2 \theta + \cos^2 \theta = 1$ (2) $\sin(90 \pm \theta)$
 (3) $\cos 4\pi = 1$ (4) all</p> <p>2. $\sin(8\pi + \theta)$ will be ;
 (1) $-\sin \theta$ (2) $\sin \theta$ (3) $\cos \theta$ (4) $-\cos \theta$</p> <p>3. If $\cos A = \frac{9}{41}$, and $\tan A$ negative, find $\tan A$?
 (1) $\frac{40}{9}$ (2) $\frac{-40}{9}$ (3) $\frac{-41}{40}$ (4) $\frac{41}{40}$</p> <p>4. Find the value of $\cos(165^\circ)$:
 (1) $\frac{1-\sqrt{3}}{2\sqrt{2}}$ (2) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
 (3) $\frac{-(\sqrt{3}-1)}{2\sqrt{2}}$ (4) $\frac{-(\sqrt{3}+1)}{2\sqrt{2}}$</p> <p>5. $2\sin x \cos x$ is equal to :
 (1) $\sin 2x$ (2) $\sin \frac{x}{2}$
 (3) $\sin x$ (4) $4 \sin x$</p> <p>6. If $y = e^{1000}$ then value of $\frac{dy}{dx}$ is :
 (1) e^{1000} (2) $1000 e^{999}$
 (3) Zero (4) 1</p> | <p>7. If $y = x^3 + 2x + 1$ then $\frac{dy}{dx}$ at $x = 1$ is :
 (1) 6 (2) 7 (3) 8 (4) 5</p> <p>8. If $y = e^x \cdot \cot x$ then $\frac{dy}{dx}$ will be :
 (1) $e^x \cot x - \operatorname{cosec}^2 x$ (2) $e^x \operatorname{cosec}^2 x$
 (3) $e^x [\cot x - \operatorname{cosec}^2 x]$ (4) $e^x \cot x$</p> <p>9. If $y = x \ln x$ then $\frac{dy}{dx}$ will be :
 (1) $\ln x + x$ (2) $1 + \ln x$
 (3) $\ln x$ (4) 1</p> <p>10. If $y = \sin x + \cos x$ then $\frac{dy}{dx}$ at $x = \pi$ is :
 (1) 1 (2) 0 (3) -1 (4) 2</p> <p>11. If $y = (x + 1)(x - 1)$ then $\frac{dy}{dx}$ at $x = 1$ is :
 (1) 1 (2) 0 (3) -1 (4) 2</p> <p>12. $y = 2u^3$, $u = 8x - 1$ then find the $\frac{dy}{dx} = ?$
 (1) $6u^2$ (2) $48u^2$ (3) 8 (4) $12u$</p> |
|---|---|

SPACE FOR ROUGH WORK

13. If $y = e^{(x^2+x)}$ then $\frac{dy}{dx} = ?$

(1) e^{x^2+x} (2) $e^{(2x+1)}$

(3) $(2x+1)e^{(x^2+x)}$ (4) 0

14. Particle's position as a function of time t is given by $x = -t^2 + 4t + 4$. Find the maximum value of position coordinates of particle.

(1) 2 (2) 8 (3) 16 (4) 12

15. Differentiation of $y = \sin 4x$ (w.r.t. x) is :

(1) $\cos 4x$ (2) $-\cos 4x$

(3) $4\cos 4x$ (4) None of these

16. If $y = x^3$ then $\frac{d^2y}{dx^2}$ is :

(1) $6x^2$ (2) $6x$ (3) $3x^2$ (4) $3x$

17. $\frac{d}{dx} \ln(\sin x) =$

(1) $\frac{1}{\sin x}$ (2) $\frac{1}{\cos x}$

(3) $\cot x$ (4) $e^{\sin x}$

18. $\frac{d}{dx} \left(\frac{1}{x^3} \right)$ is equal to

(1) $\frac{-3}{x^4}$ (2) $\frac{x^2}{3}$ (3) $\frac{-x^2}{3}$ (4) $3x^2$

19. $\int x^2$ is equal to

(1) $\frac{x^3}{3} + c$ (2) $2x + c$

(3) $\frac{2x^3}{3} + c$ (4) Meaningless

20. $\int (3t^2 - 2t) dt =$

(1) $t^2 - t^3 + c$ (2) $t^3 - t^2 + c$

(3) $t^3 + t^2 + c$ (4) None of these

21. $\int_{-\pi/2}^{\pi/2} \cos x \, dx =$

(1) 1 (2) 3 (3) 2 (4) 5

22. $\int_0^{\pi/6} \sec^2 x \, dx =$

(1) $\frac{1}{3}$ (2) $\sqrt{3}$ (3) $\frac{1}{\sqrt{3}}$ (4) 3

23. If $\tan \theta = \frac{4}{3}$ then $\cos \theta$ will be :

(1) $\frac{3}{5}$ (2) $\frac{4}{5}$ (3) $\frac{2\sqrt{2}}{3}$ (4) $\frac{3}{4}$

SPACE FOR ROUGH WORK

24. $\sin(75^\circ)$ is equal to

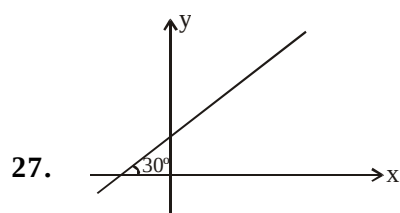
- (1) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (2) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
(3) $\frac{\sqrt{3}-1}{\sqrt{2}}$ (4) $\frac{\sqrt{3}+1}{\sqrt{2}}$

25. $\sin 15^\circ \cos 15^\circ$ is equal to

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$
(3) $\frac{1}{4}$ (4) $\frac{1}{12}$

26. $1 + \cot^2 \theta$ is equal to

- (1) $\tan^2 \theta$ (2) $\sec^2 \theta$
(3) $\cos^2 \theta$ (4) $\operatorname{cosec}^2 \theta$



Slope of line shown in the figure is :

- (1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$
(3) 3 (4) $\frac{1}{3}$

28. If $y = x + \frac{1}{x}$ then $\frac{dy}{dx}$ is :

- (1) $1 + \frac{1}{x^2}$ (2) $x - \frac{1}{x^2}$
(3) $1 - \frac{1}{x^2}$ (4) $x^2 - \frac{1}{x^2}$

29. If $y = e^{-x}$ then $\frac{dy}{dx}$ is :

- (1) e^{-x} (2) $-e^{-x}$ (3) $\frac{1}{e^x}$ (4) e^{-2x}

30. If $y = e^x \sin x$ then $\frac{dy}{dx}$ is :

- (1) $e^x (\sin x - \cos x)$ (2) $e^{-x} (\sin x - \cos x)$
(3) $e^x (\sin x + \cos x)$ (4) $e^x (\cos x - \sin x)$

31. If $y = \sin x^2$ then $\frac{dy}{dx}$ is :

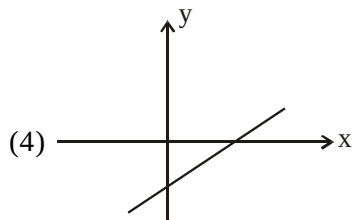
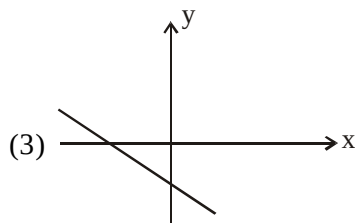
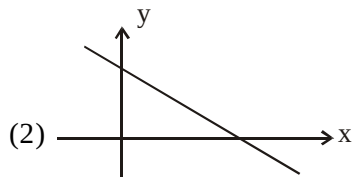
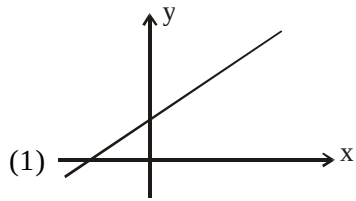
- (1) $\cos x^2$ (2) $\cos (2x)$
(3) $2x \cos x^2$ (4) $x^2 \cos x^2$

32. If $y = x^{1/2}$ then $\frac{d^2y}{dx^2}$ is :

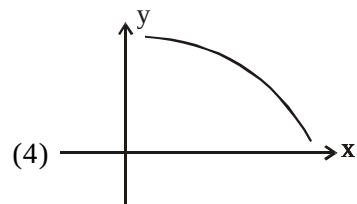
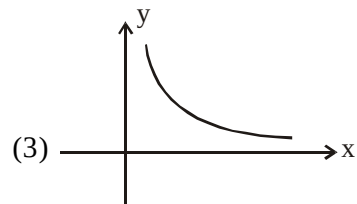
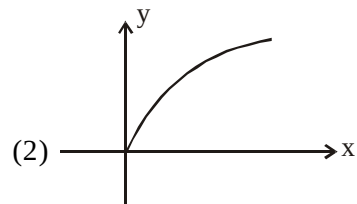
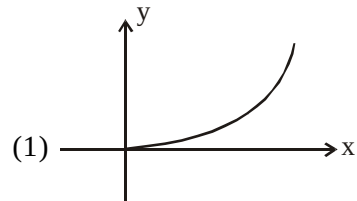
- (1) $\frac{x^{1/2}}{2}$ (2) $\frac{-1}{4x^{3/2}}$
(3) $\frac{-1}{4x^2}$ (4) $\frac{-1}{4} x^{3/2}$

SPACE FOR ROUGH WORK

33. Correct graph for $y = -3x + 2$ is :



34. Correct graph for $xy = \text{const}$ may be :



SPACE FOR ROUGH WORK

35. If $y = \sin x + \cos x$ find $\frac{dy}{dx}$

- (1) $\sin x - \cos x$ (2) $-\sin x + \cos x$
(3) $\sin x + \cos x$ (4) $-\sin x - \cos x$

36. $y = x^\alpha + x^\beta$ find $\frac{dy}{dx}$

- (1) $\alpha x^{\alpha-1} + \beta x^{\beta-1}$ (2) $x^{\alpha-1} + x^{\beta-1}$
(3) $\frac{x^{\alpha-1}}{\alpha} + \frac{x^{\beta-1}}{\beta}$ (4) $\frac{1}{x^\alpha} + \frac{1}{x^\beta}$

37. $\sin 37^\circ \times \cos 53^\circ$ equal to

- (1) $\frac{12}{25}$ (2) $\frac{9}{25}$ (3) $\frac{16}{25}$ (4) $\frac{25}{25}$

38. $\sec(330^\circ)$ equal to

- (1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{2}$ (3) $\frac{2}{\sqrt{3}}$ (4) $\frac{-2}{\sqrt{3}}$

39. $\tan(210^\circ)$ equals to

- (1) $\frac{1}{\sqrt{3}}$ (2) $\sqrt{3}$ (3) $-\sqrt{3}$ (4) $-\frac{1}{\sqrt{3}}$

40. If $f(x) = \sin x + \cos x$ find value of $f(30^\circ)$

- (1) $\frac{1}{2} + \frac{\sqrt{3}}{2}$ (2) $\frac{1}{2} - \frac{\sqrt{3}}{2}$

(3) $\frac{\sqrt{3}}{2} - \frac{1}{2}$

(4) $\frac{-\sqrt{3}}{2} - \frac{1}{2}$

41. $y = -\frac{1}{\sqrt{3}}x + 5$ find the intercept made by this

line on y-axis :

- (1) 5 (2) $\sqrt{3}$ (3) $5\sqrt{3}$ (4) 3

42. $y = x^3 + 2x^2 + 7x + 8$ then $\frac{dy}{dx}$ will be :

- (1) $3x^2 + 2x + 15$ (2) $3x^2 + 4x + 7$
(3) $x^3 + 2x^2 + 15$ (4) $x^3 + 4x + 7$

43. If $y = \sin^2 x + \cos^2 x$ then $\frac{dy}{dx}$ is :

- (1) $2\sin x + 2\cos x$ (2) $\sin^2 x$
(3) $+2\sin x - 2\cos x$ (4) Zero

44. If $y = a \log_e x$ then $\frac{dy}{dx}$ is :

- (1) $\frac{x^{a-1}}{a-1}$ (2) $\frac{a}{x}$ (3) $\frac{x}{a}$ (4) ax^{a-1}

45. Sum of $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$ is :

- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{3}{2}$

SPACE FOR ROUGH WORK

PART-II [CHEMISTRY]

- | | |
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| <p>46. Chemical equation is balanced according to the law of :
 (1) Multiple proportion (2) Reciprocal proportion
 (3) Conservation of mass (4) Definite proportion</p> <p>47. Carbon and oxygen combine to form two oxides, carbon monoxide and carbon dioxide in which the ratio of the weights of carbon and oxygen is respectively 12 : 16 and 12 : 32. These figures illustrate the :
 (1) Law of multiple proportions
 (2) Law of reciprocal proportions
 (3) Law of conservation of mass
 (4) Law of constant proportions</p> <p>48. A sample of pure carbon dioxide, irrespective of its source contains 27.27% carbon and 72.73% oxygen. The data support.
 (1) Law of constant composition
 (2) Law of conservation of mass
 (3) Law of reciprocal proportions
 (4) Law of multiple proportions</p> <p>49. n g of substance X reacts with m g of substance Y to form p g of substance R and q g of substance S. This reaction can be represented as, $X + Y = R + S$. The relation which can be established in the amounts of the reactants and the products will be :
 (1) $n - m = p - q$ (2) $n - q = p - m$
 (3) $n + p = m + q$ (4) $n + q = m + p$</p> | <p>50. The modern atomic weight scale is based on :
 (1) C^{12} (2) O^{16} (3) H^1 (4) C^{13}</p> <p>51. 1 amu is equal to :
 (1) $\frac{1}{12}$ of C-12 (2) $\frac{1}{14}$ of O-16
 (3) 1 g of H_2 (4) 1.66×10^{-23} kg</p> <p>52. 0.44 g of a colourless oxide of nitrogen occupies 224 mL at STP. The compound is :
 (1) N_2O (2) NO
 (3) N_2O_2 (4) NO_2</p> <p>53. How many atoms are contained in one mole of sucrose ($C_{12}H_{22}O_{11}$) ?
 (1) $45 \times 6.02 \times 10^{23}$ atom / mole
 (2) $20 \times 6.02 \times 10^{23}$ atom / mole
 (3) $5 \times 6.02 \times 10^{23}$ atom / mole
 (4) $60 \times 6.02 \times 10^{23}$ atom / mole</p> <p>54. 4 g-atom of Ag weighs : (Ag = 108)
 (1) 108 g (2) 4 g (3) 432 g (4) 216 g</p> <p>55. The total number of protons, electrons and neutrons is 12 g of $^{12}_6C$ is :
 (1) 1.084×10^{25} (2) 6.022×10^{23}
 (3) 6.022×10^{22} (4) 18</p> <p>56. The least number of molecules are present in :
 (1) 2 g hydrogen (2) 8 g oxygen
 (3) 4 g nitrogen (4) 16 g CO_2</p> |
|---|--|

SPACE FOR ROUGH WORK

57. Which has the maximum number of atoms ?
 (1) 24 g C(12) (2) 56 g Fe(56)
 (3) 27 g Al(27) (4) 108 g Ag(108)
58. An oxide of sulphur contains 50% S. What will be its empirical formula ?
 (1) SO (2) SO₂ (3) SO₃ (4) S₂O₃
59. 1000 g aqueous solution of CaCO₃ contains 10g of calcium carbonate. Concentration of solution is :
 (1) 10 ppm (2) 100 ppm
 (3) 1000 ppm (4) 10000 ppm
60. A gaseous mixture contains 8 g of Helium and 22.4 litre of oxygen at STP. What is the mole fraction of oxygen in the mixture.
 (1) 22.4/(22.4 + 44.8) (2) 2/3
 (3) 22.4/(22.4 + 22.4) (4) 3/4
61. The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is :
 (1) 8/5 (2) 0.5 (3) 0.25 (4) 1.0
62. The number of mole present in 2 litre of 0.5 M NaOH is :
 (1) 2 (2) 1 (3) 0.1 (4) 0.5
63. 9.8 g of HSO₄ is present in 2 litre of solution. The molarity of the solution is :
 (1) 0.1 M (2) 0.05 M
 (3) 0.01 M (4) 0.2 M
64. 6.02×10^{20} molecules of urea are present in 100 mL solution. The concentration of urea solution is :
 (1) 0.1 M (2) 0.01 M
 (3) 0.02 M (4) 0.001 M
65. The solutions A and B are 0.1 molar and 0.2 molar of a substance. If 100 mL of A are mixed with 25 mL of B and there is no change in volume, then the final molarity of the solution is :
 (1) 0.15 M (2) 0.18 M
 (3) 0.12 M (4) 0.30 M
66. In a gaseous reaction of the type
 $aA + bB \longrightarrow cC + dD$
 which statement is wrong ?
 (1) a litre of A combines with b litre of B to give C and D
 (2) a mole of A combines with b moles of B to give C and D
 (3) a g of A combines with b g of B to give C and D
 (4) a molecules of A combines with b molecules of B to give C and D
67. 9 g of Al will react, with

$$2Al + \frac{3}{2}O_2 \rightarrow Al_2O_3$$

 (1) 6g O₂ (2) 8 g O₂
 (3) 9 g O₂ (4) 4 g O₂

SPACE FOR ROUGH WORK

68. The number of litres of air required to burn 8 litres of C_2H_2 is - (Given : $V(O_2) = 20\%$ of $V(\text{air})$)
(1) 40 (2) 60 (3) 80 (4) 100
69. If 2.5 mole of $BaCl_2$ is mixed with 1 mole of Na_3PO_4 the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is :
 $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
(1) 0.7 (2) 0.5 (3) 0.3 (4) 0.1
70. 4 g of hydrogen are ignited with 8 g of oxygen. The weight of water formed is -
(1) 0.5 g (2) 9 g
(3) 4.5 g (4) 2.5 g
71. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gives C, 38.71% and H, 9.67%. The empirical formula of the compound would be :
(1) CHO (2) CH_4O
(3) CH_3O (4) CH_2O
72. The number of water molecules is maximum in :-
(1) 18 gram of water
(2) 18 moles of water
(3) 18 molecules of water
(4) 1.8 g-molecule of water
73. The ratio of masses of oxygen and Helium in a particular gaseous mixture is 1 : 4. The ratio of number of molecules is :
(1) 1 : 8 (2) 3 : 16
(3) 1 : 4 (4) 1 : 32
74. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide.
(1) M_2O (2) M_2O_3
(3) MO (4) M_3O_4
75. 12 L of H_2 and 11.2 L of Cl_2 are mixed and reacted to give HCl. The composition by volume of mixture is :
(1) 24 L of HCl (g)
(2) 0.8 L Cl_2 and 20.8 L HCl (g)
(3) 0.8 L H_2 and 22.4 L HCl (g)
(4) 22.4 L HCl (g)
76. Number of atoms in 4.25 g of NH_3 is :
(1) 6.023×10^{23} (2) $4 \times 6.023 \times 10^{23}$
(3) 1.7×10^{24} (4) $4.5 \times 6.023 \times 10^{23}$
77. Mass of 5 moles of $C_6H_{12}O_6$ is :
(1) 900 amu (2) 1.5×10^{-21} gm
(3) 4.5×10^{-21} gm (4) Both (1) & (2)
78. A_1 g of an element gives A_2 g of its oxide. The equivalent mass of the element is :
(1) $\frac{A_2 - A_1}{A_1} \times 8$ (2) $\frac{A_2 - A_1}{A_2} \times 8$
(3) $\frac{A_1}{A_2 - A_1} \times 8$ (4) $\frac{A_2 - A_1}{A_2} \times 35.5$

SPACE FOR ROUGH WORK

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| <p>79. When the specific heat of a metallic element is 0.214 cal g^{-1}, the atomic mass will be closest to which one of the following?</p> <p>(1) 1 (2) 12
(3) 30 (4) 66</p> <p>80. 0.32 g of a metal gave on treatment with an acid 112 mL of hydrogen at NTP. Calculate equivalent mass of the metal :</p> <p>(1) 58 (2) 32 (3) 11.2 (4) 24</p> <p>81. 74.5 g of a metallic chloride contains 35.5 g of chlorine. The equivalent mass of the metal is :</p> <p>(1) 19.5 (2) 35.5
(3) 39.0 (4) 78.0</p> <p>82. The oxide of an element possesses the molecular formula, M_2O_3. If the equivalent mass of the metal is 9, the atomic mass of the metal will be :</p> <p>(1) 27 (2) 18 (3) 9 (4) 4.5</p> <p>83. The atomic mass of an element is 27. If valency is 3, the vapour density of the volatile chloride will be :</p> <p>(1) 66.75 (2) 6.675 (3) 667.5 (4) 81</p> <p>84. The largest number of molecules is in :</p> <p>(1) 28 g of CO (2) 46 g of $\text{C}_2\text{H}_5\text{OH}$
(3) 36 g of H_2O (4) 54 g of N_2O_5</p> | <p>85. The number of grams of H_2SO_4 present in 0.25 mole of H_2SO_4 is :</p> <p>(1) 0.245 (2) 2.45
(3) 24.5 (4) 49.0</p> <p>86. The number of molecules in 89.6 litre of a gas at NTP are :</p> <p>(1) 6.02×10^{23} (2) $2 \times 6.02 \times 10^{23}$
(3) $3 \times 6.02 \times 10^{23}$ (4) $4 \times 6.02 \times 10^{23}$</p> <p>87. 19.7 kg of gold was recovered from a smuggler. The atoms of gold recovered are : ($\text{Au} = 197$)</p> <p>(1) 100 (2) 6.02×10^{23}
(3) 6.02×10^{24} (4) 6.02×10^{25}</p> <p>88. Four flasks of 1 litre capacity each are separately filled with gases H_2, He, O_2 and O_3. At the same temperature and pressure the ratio of the number of atoms of these gases present in different flasks would be :</p> <p>(1) 1 : 1 : 1 : 1 (2) 2 : 1 : 2 : 3
(3) 1 : 2 : 1 : 3 (4) 3 : 2 : 2 : 1</p> <p>89. The percentage of nitrogen in urea (NH_2CONH_2), is :</p> <p>(1) 38.4 (2) 46.6
(3) 59.1 (4) 61.3</p> <p>90. One micro gram is equal to :</p> <p>(1) 10^{-3} g (2) 10^3 g
(3) 10^6 g (4) 10^{-6} g</p> |
|---|--|

SPACE FOR ROUGH WORK

PART-III [BIOLOGY]

- | | |
|--|--|
| <p>91. Cell was first discovered and named by _____ in _____. Hooke's description of these cells was published in _____.
 (1) Leeuwenhoek, 1674, species plantarum
 (2) Robert hooke, 1661, origin of cell
 (3) Robert hooke, 1665, micrographia
 (4) Robert brown, 1831, cellula</p> <p>92. Which is/are prokaryotic cell?
 (1) BGA (2) bacteria
 (3) PPLO (4) all</p> <p>93. Extra chromosomal DNA occurs in
 (1) Mitochondria (2) Chloroplast
 (3) Bacteria (4) All</p> <p>94. Find out wrong pair
 (1) simple microscope – Leeuwenhoek
 (2) compound microspe - Zacharias Jansen
 (3) electron microscope – knoll and ruska
 (4) cell lineage theory–schleiden and schwann</p> <p>95. Which of the following is exception of cell theory?
 (1) Virus
 (2) Multicellular organism
 (3) Bacteria
 (4) All</p> | <p>96. Diameter of human RBC
 (1) 5 micron (2) 7 micron
 (3) 9 micron (4) 11 micron</p> <p>97. Which of the following statement is incorrect?
 (1) Leeuwenhoek saw bacteria for the first time
 (2) Electron microscope was invented by knoll and ruska
 (3) Veretbrates have nucleated RBC
 (4) Prokaryotes have naked single stranded circular DNA</p> <p>98. Meristematic cells have
 (1) Primary cell wall
 (2) Secondary cell wall
 (3) Tertiary cell wall
 (4) All</p> <p>99. What is true about mesosomes?
 (1) Help in cell wall formation
 (2) Help in genome duplication
 (3) Help in respiration
 (4) All</p> <p>100. Selective permeable membrane allow
 (1) All materials
 (2) Only certain materials
 (3) Only ions
 (4) Only water</p> |
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SPACE FOR ROUGH WORK

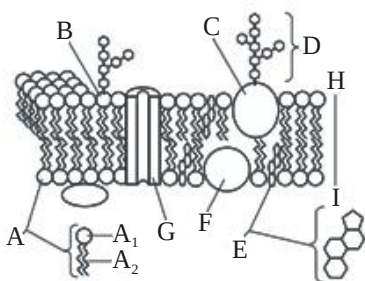
101. Which pair is incorrect?

(1)	Phospholipids	Main fabric of the membrane
(2)	Cholesterol	Tucked between the hydrophobic tails of the membrane phospholipids
(3)	Integral proteins	Embedded in the phospholipid bilayer ; may or may not extend through both layers
(4)	Peripheral proteins	On the inner or outer surface of the phospholipid bilayer, embedded in its hydrophobic core

102. Quantasomes occur in

- (1) Stroma (2) Grana
(3) Mitochondria (4) Lipochondria

103. Which option is wrong regarding ultrastructure of cell membrane.



- (1) A-phospholipid, G-channel protein, D-glycoprotein, F-integral protein
(2) A₁-hydrophilic head, F- intrinsic protein, E- cholesterol
(3) B-phosphoglycerides, D-glycoprotein, G-channel protein
(4) C-intrinsic protein, E-extrinsic protein, A₂-hydrophobic tail

104. Nucleolemma derive from

- (1) SER (2) RER
(3) Golgi body (4) Lysosome

105. Glycosidation of proteins and lipids occur in _____.

- (1) Plastids (2) Vacuoles
(3) Golgi apparatus (4) Lysosomes

106. Which statement is wrong?

- (1) The cytosol shows Tyndal effect (light scattering by particle in colloidal solution) and Brownian motion (random moving of particles)
(2) in the cell K⁺ and Mg⁺⁺ can be high, and Na⁺ and Cl⁻ high outside the cell.
(3) human cytosolic pH lies between 8.0-8.4,
(4) The cytoplasm is everything within the cell except for the nucleus.

107. Which one is non-membranous organelle

- (1) Centriole (2) Ribosome
(3) Nucleolus (4) All

108. Which of the following pair is incorrect?

- (1) Basal granule-centriole like structure
(2) Axoneme – core of cilia
(3) Radial spokes-connect two peripheral tubules
(4) Centriole – 9 + 0

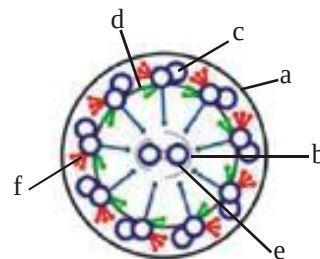
SPACE FOR ROUGH WORK

- 109.** Which of the following statements about the lipid constituents of membranes is incorrect?
- (1) Flipflop movement of membrane lipids is catalyzed by special proteins.
 - (2) Protein can diffuse laterally through the membrane
 - (3) Lateral movement of membrane lipids occurs rapidly within the plane of the bilayer.
 - (4) Transverse movement of membrane lipids within the bilayer is unrestricted.
- 110.** Large amoeboid-type cells, called macrophages, remove bacteria and worn-out red blood cells by a process called ____.
- (1) Pinocytosis
 - (2) Phagocytosis
 - (3) Endocytosis
 - (4) Both (1) and (2)
- 111.** Lipid-soluble molecules and gases enter the cell by ____.
- (1) Diffusion through the channel proteins
 - (2) Diffusion through the lipid bilayer
 - (3) Osmosis through the lipid bilayer
 - (4) All options are correct
- 112.** Why are proteins involved in active transport often called "pumps"?
- (1) They use energy to move a substance against its concentration gradient.
 - (2) They use energy to move a substance with its concentration gradient.
 - (3) They use energy to bind the substance to the carrier.
 - (4) They cause the plasma membrane to invaginate like a pump that pulls up water against the force of gravity

- 113.** The protein storing leucoplasts are called ____.

- (1) Amyloplast
- (2) Elaioplast
- (3) Aleuroplast
- (4) Mitoplast

- 114.** Identify the figure and find the correct option



- (1) a-cell membrane, e- interdoublet bridge, c- peripheral doublet, f-dynein arms
 - (2) c-peripheral doublet e-linker, d-nexin, a-cell membrane
 - (3) a-cell membrane, c-linker, d-dynein arms, f-radial spokes
 - (4) b-central sheath, c- peripheral doublet, e-central singlet, f-dynein arm
- 115.** Which of the following is the main function of intermediate filaments?
- (1) To provide tracks for guiding intracellular transport
 - (2) To enable cells to crawl
 - (3) To enable cells to withstand the mechanical stress that occurs when cells are stretched
 - (4) help in cell division

SPACE FOR ROUGH WORK

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| <p>116. The primary structural component of centrioles are :</p> <p>(1) Microtubules
(2) Microfilaments
(3) Intermediate filaments
(4) Basal bodies</p> <p>117. Mitochondria and chloroplast are</p> <p>(i) Semiautonomous organelle
(ii) Endosymbionts
(iii) Derive from pre-existing organelle
(iv) They do not have protein synthesis machinery</p> <p>(1) i & ii (2) i, ii & iii
(3) i, iii & iv (4) i, ii, iii & iv</p> <p>118. The type of plastids which help in pollination is</p> <p>(1) Leucoplast (2) Bioplast
(3) Chloroplast (4) Chromoplast</p> <p>119. Which is a feature of plasmodesmata</p> <p>(1) Channel between cells
(2) Occur in all cells
(3) Lined by cell membrane and desmotubules
(4) Both (1) and (3)</p> <p>120. Which one is not a eukaryote?</p> <p>(1) Acetabularia (2) Euglena
(3) Nostoc (4) Yeast</p> <p>121. Membranous covering of vacuole is called</p> <p>(1) Tonoplast (2) Biomembrane
(3) Permeable (4) None</p> | <p>122. All are functions of cytoskeleton except</p> <p>(1) Mobility of cell
(2) Transport of substances
(3) Cell shape
(4) Support to organelles</p> <p>123. gas vacuoles are found in</p> <p>(1) Mycoplasma (2) Yeast
(3) Bacteria (4) All</p> <p>124. elementary particles are found in/on</p> <p>(1) Inner membrane
(2) Outer membrane
(3) Perimitochondrial space
(4) Matrix</p> <p>125. Flat membrane bound sac like structure present in stroma of chloroplast are</p> <p>(1) Grana (2) Cisterna
(3) Fret lamellae (4) Thylakoids</p> <p>126. Centrioles and centrosomes occur in the cells of</p> <p>(1) Animals (2) Green plants
(3) Cyanobacteria (4) Both (1) and (3)</p> <p>127. _____ is a polymorphic organelle.</p> <p>(1) Golgi body (2) Lysosome
(3) ER (4) Ribosome</p> <p>128. Proteins which are synthesized by RER, usually</p> <p>(1) For internal regulation
(2) For internal storage
(3) Exported from the cell
(4) To digest food in lysosomes</p> |
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SPACE FOR ROUGH WORK

129. Self replicating organelles are

- (1) Centrioles (2) Mitochondria
(3) Chloroplast (4) All

130. Average thickness of plasma membrane :

- (1) 40 Å (2) 70 Å
(3) 150 Å (4) 200 Å

131. Which of the following is not associated with lysosomes

- (1) Crinophagy (2) Autophagy
(3) Secretion (4) Heterophagy

132. Water soluble pigments are found in

- (1) Chromoplast (2) Vacuole
(3) Mitochondria (4) All

133. Which is not present in a plant cell

- (1) Lysosome (2) Centriole
(3) Ribosome (4) Both (1) and (2)

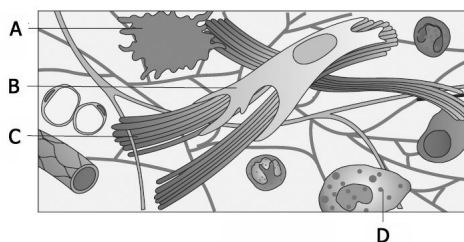
134. Centrosomes are also known as

- (1) Nuclear organizer center
(2) Microtubule organizing center
(3) Chromosomal organizing center
(4) All

135. microfilaments are mainly composed of

- (1) Actin (2) Myosin
(3) Tubulin (4) Chitin

136. Identify A, B C & D in the given diagram of areolar tissue :



- (1) A=Mast cell; B=Fibroblast; C=Collagen fibres; D=Macrophage
(2) A=Macrophage; B=Mast cell ; C=Collagen fibres; D=Fibroblast
(3) A=Macrophage; B=Fibroblast; C=Mast cell; D=Collagen fibres
(4) A=Macrophage; B=Fibroblast; C=Collagen fibres; D=mast cell

137. Histamine secreting cells are found in:

- (1) Connective tissues (2) Epithelium tissue
(3) Muscular tissue (4) Nervous tissue

138. Component of blood responsible for producing antibodies is:

- (1) Thrombocytes (2) Monocytes
(3) Erythrocytes (4) Lymphocytes.

139. Basement membrane is made up of:

- (1) Product of epithelial cell
(2) Epidermal cell only
(3) Endodermal cell only
(4) Both (2) and (3)

SPACE FOR ROUGH WORK

140. Protein present in cartilage is:

- (1) Cartilagin (2) Ossein
(3) Chondrin (4) None of these.

141. Tendon is made up of:

- (1) Yellow fibrous connective tissue
(2) Modified white fibrous tissue
(3) Areolar tissue
(4) Adipose tissue

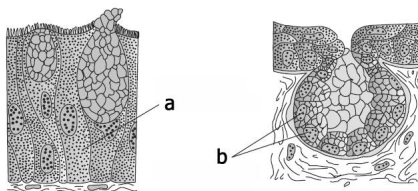
142. Ligament is a/an:

- (1) Inelastic white fibrous tissue
(2) Modified white fibrous tissue
(3) Modified yellow elastic fibrous tissue
(4) None of the above.

143. Characteristic of simple epithelium is that they:

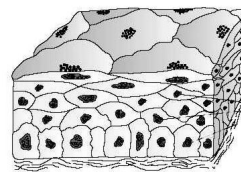
- (1) Form lining of body cavities
(2) Has protective function
(3) Has limited role in secretion and absorption
(4) All of the above.

144. Identify 'a' & 'b' in the given diagrams and select the correct answer :



- (1) 'a' is unicellular and 'b' is multicellular gland
(2) Sudoriferous glands are example of 'b'
(3) Paneth cells of intestine are example of 'a'
(4) All of the above

145. Identify the given figure :



- (1) Simple epithelium
(2) Areolar tissue
(3) Compound epithelium
(4) Adipose tissue

146. During an injury nasal septum gets damaged and for its recovery which cartilage is preferred

- (1) Elastic cartilage (2) Hyaline cartilage
(3) Calcified cartilage (4) Fibrous cartilage.

147. Collagen is:

- (1) Fibrous protein (2) Globular protein
(3) Lipid (4) Carbohydrate.

148. Which one of the following contains the largest quantity of extracellular material?

- (1) Striated muscle
(2) Areolar tissue
(3) Stratified epithelium
(4) Myelinated nerve fibres

149. Mast cells of connective tissue contain:

- (1) Vasopressin and relaxin
(2) Heparin and histamine
(3) Heparin and calcitonin
(4) Serotonin and melanin.

SPACE FOR ROUGH WORK

150. Four healthy people in their twenties got involved in injuries resulting in damage and death of few cells of the following. Which of the cells are least likely to be replaced by new cells ?

- (1) Liver cells
- (2) Neurons
- (3) Malpighian layer of the skin
- (4) Osteocytes

151. Areolar connective tissue joins:

- (1) Bones with bones
- (2) Fat body with bones
- (3) Integument with muscles
- (4) Bones with muscles

152. Characteristic of epithelial tissue is :

- (1) Never produce glands
- (2) Cells can undergo rapid divisions
- (3) Abundant vascularisation
- (4) Large intercellular spaces

153. Desmosomes -

- (1) Connect the epithelial cells
- (2) Types of lysosomes
- (3) Granules of muscle fibres
- (4) Found in bone

154. Gap junctions :

- (1) Help to stop substances from leaking across a tissue
- (2) Perform cementing to keep neighbouring cells together
- (3) Provide stretchability to the epithelium
- (4) Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells

155. In stratified squamous epithelium :

- (1) Outer most layer squamous & inner most is cuboidal
- (2) Outer most layer cuboidal & inner most is squamous.
- (3) Outermost layer columnar & inner most is cuboidal
- (4) Outermost layer is cuboidal & innermost is columnar

156. Tiles like epithelium is present in :

- (1) Ependymal membrane
- (2) Endothelium
- (3) Schneiderian membrane
- (4) None of these

157. Lining of larynx is :

- (1) Stratified ciliated columnar epithelium
- (2) Stratified squamous epithelium
- (3) Stratified cuboidal epithelium
- (4) Stratified non-ciliated columnar epithelium

SPACE FOR ROUGH WORK

158. Epithelial lining of cornea is composed of :

- (1) Transitional
- (2) Stratified squamous non-keratinised
- (3) Simple Cuboidal
- (4) Simple Squamous

159. The covering of articular cartilage is made up of :

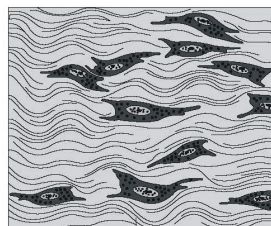
- (1) Areolar CT (2) Yellow fibrous CT
- (3) White fibrous CT (4) Reticular CT

160. Identify the locations where compound epithelium is found :

- (A) Surface of the skin
- (B) Moist surface of buccal cavity
- (C) Alveoli of lungs
- (D) Wall of blood vessels
- (E) PCT
- (F) Pharynx

- (1) A, B, C (2) A, B, F
- (3) B, C, E (4) B, C, F

161. Consider the following figures & choose the correct answer :



(a)



(b)

- (1) Tendons & ligaments are example of 'a'
- (2) 'a' is irregular connective tissue & 'b' is regular connective tissue
- (3) Cartilage is an example of 'a'
- (4) Both a & b are example of loose connective tissue

162. Mammary glands of human are modified :

- (1) Sweat gland (2) Sebaceous gland
- (3) Lacrymal gland (4) Endocrine gland

163. Identify the wrong statement :

- (1) Loose connective tissue has cells & fibres loosely arranged in a semi fluid ground substance
- (2) The excess of nutrients which are not used immediately are converted into fats & are stored in areolar tissue
- (3) Areolar & adipose connective tissue are example of loose connective tissue
- (4) Adipose tissue is present beneath the skin

164. Name the tissue that forms glands :

- (1) Epithelium (2) Muscular
- (3) Connective (4) Nervous

165. The cell junctions called tight, adhering and gap junctions are present in which tissue :

- (1) Muscular (2) Connective
- (3) Neural (4) None

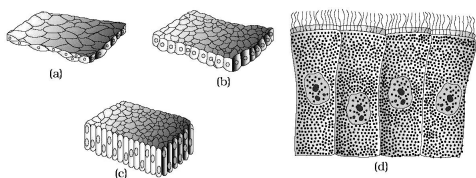
SPACE FOR ROUGH WORK

- 166.** Consider the following statement given below
- (A) It forms the lining of lungs alveolar cavities
(B) It forms lining of oesophagus
(C) It forms lining of salivary glands and sweat glands
(D) It form the covering of muscle bundle
- Which statements associated with epithelial tissue

- (1) A, B & C (2) A, B & D
(3) B, C & D (4) All

- 167.** Simple squamous epithelial tissue is present in
- (1) Blood vessels & Alveoli
(2) Ducts of gland & Tubular parts of Nephrons
(3) In PCT & Fallopian tube
(4) lining of stomach and Intestine

- 168.** Identify a, b, c & d and choose the correct answer :



- (1) a = Flattened cells, b = Cuboidal cells,
c = Columnar cells, d = Ciliated columnar cells
(2) a = Cuboidal cells, b = Flattened cells,
c = Columnar cells, d = Ciliated columnar cells

- (3) a = Ciliated columnar cells, b = Columnar cells, c = Flattened cells, d = Cuboidal cells
(4) a = Columnar cells, b = Ciliated columnar cells, c = Cuboidal cells, d = Flattened cells

- 169.** Ciliated epithelium present in :
- (A) Inner surface of hollow organs like of Bronchioles
(B) Inner surface of Fallopian tube
(C) Inner surface of pancreatic duct
(D) Inner surface of Duodenum

Choose correct options

- (1) A & C (2) A & B
(3) A,B,C (4) ABCD

- 170.** Goblet cells of Alimentary canal is example of
- (1) Unicellular
(2) Multicellular
(3) Unicellular and multicellular
(4) None of these

- 171.** Tight Junctions helps to :
- (1) Stop substances from leaking across a tissue
(2) facilitate the cells to communicate with other cells
(3) Rapid transfer of ions & small molecules & big molecules
(4) Cementing to keep neighbour cells together

SPACE FOR ROUGH WORK

172. Which connective tissue not secret fibres of structural proteins :

- (1) Cartilage (2) Bone
(3) Adipose (4) Blood

173. The most abundant tissue in the body is :

- (1) Neurons (2) Muscular
(3) Connective (4) Epithelial

174. Match the column :

Column-I

Column - II

- (a) Squamous epithelium 1. Below skin
(b) Hyaline cartilage 2. Bowman's capsule
(c) Adipose tissue 3. Ear wax
(d) Cerumin 4. Trachea
5. Ducts of glands

- (1) a - 5, b - 4, c - 1, d - 3
(2) a - b, b - 1, c - 4, d - 2
(3) a - 2, b - 4, c - 1, d - 3
(4) a - 2, b - 1, c - 4, d - 3

175. Which is not the property of yellow fibres :

- (1) Contain elastin
(2) Fewer in number
(3) Straight & branched
(4) Provide toughness & strength

176. Origin of epithelial tissue is :

- (1) Ectodermal (2) Mesodermal
(3) Endodermal (4) All the above

177. Which are the examples of specialized connective tissue

- (1) Cartilage & areolar
(2) Bone and blood
(3) Areolar & adipose
(4) All of these

178. Secretion of exocrine gland include :

- (1) Mucus (2) Ear wax
(3) Both 1 & 2 (4) Hormone

179. Which is not included in connective tissue :

- (1) Cartilage (2) Adipose tissue
(3) Blood (4) Muscle

180. Diapedesis means :

- (1) Formation of WBC
(2) Formation of RBC
(3) Process by lymphocyte squeezes through thin capillary wall
(4) Movement of food in gut

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